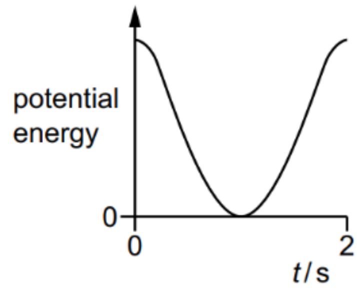
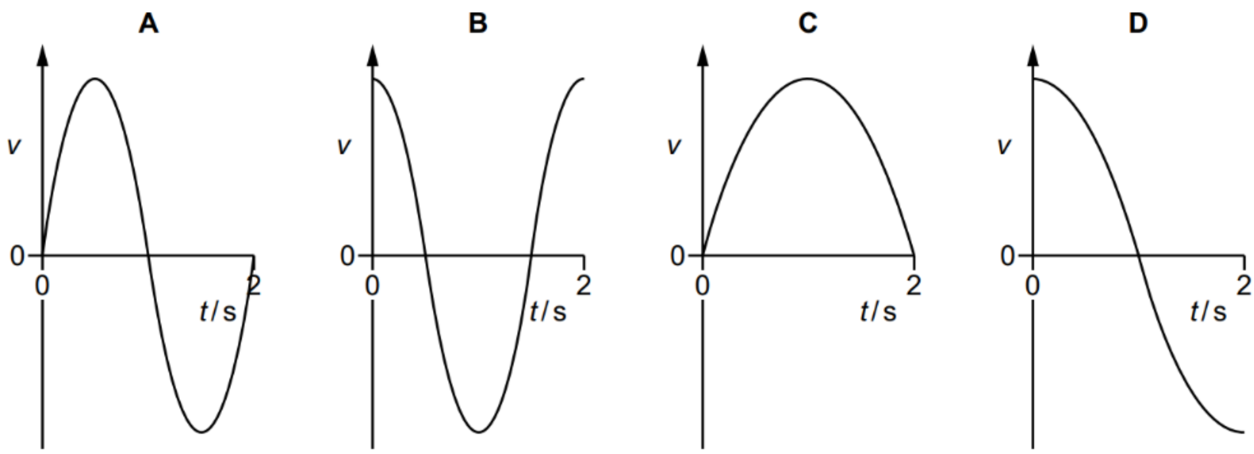
16 A particle oscillates with simple harmonic motion. The graph shows the variation, with time t, of the potential energy of the particle from $t = 0$ to $t = 2$ s.



Which graph could represent the variation, with time t, of the velocity v of the particle from $t = 0$ to $t = 2$ s?



Ans: C

Observe that the given PE graph is drawn for half a period only as there are only 2 peaks seen.

$PE \propto \cos^2 \omega t$

$KE \propto \sin^2 \omega t$

$v \propto \sin \omega t \Rightarrow v = v_0 \sin \omega t$

Option C gives the v-t graph is drawn for half a period only.